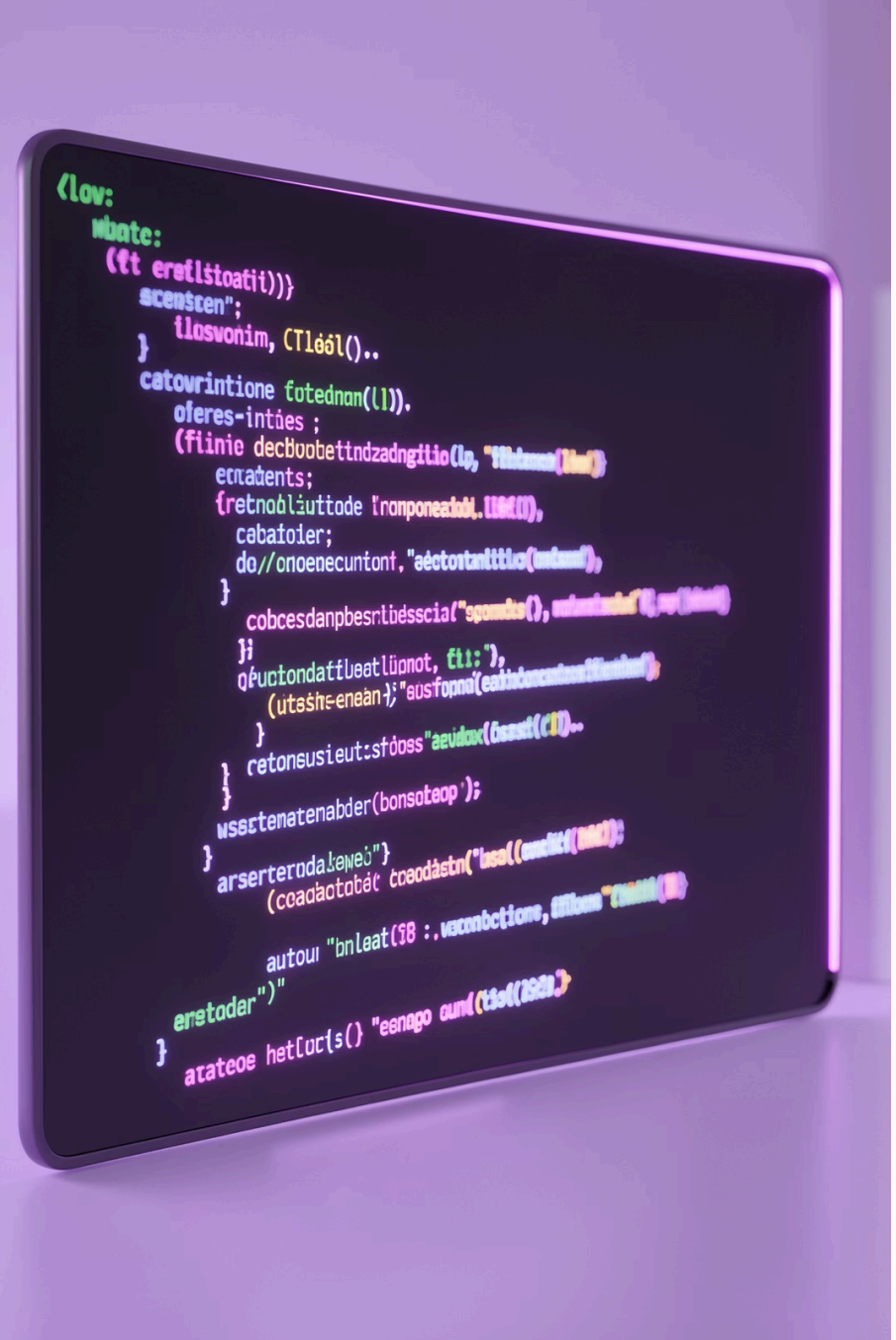

The Logic of Code

Operators, Conditions, and Loops — the three pillars that give programs their power to calculate, decide, and repeat.

PROGRAMMING FUNDAMENTALS

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Building Blocks of Logic: Operators

Operators are the workhorses of any program — they perform operations on data, compare values, and combine conditions to drive logic.

Arithmetic

`+ - * / %`

The math behind the code — addition, subtraction, multiplication, division, and modulo.

Comparison

`== != > < >= <=`

Evaluate two values and return `true` or `false` — the foundation of decisions.

Logical

`&& || !`

AND, OR, and NOT — combine multiple conditions into powerful expressions.

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Making Decisions: Conditional Statements

The `if` Statement

The most fundamental decision-maker in code. If a condition is `true`, the block executes — otherwise it's skipped.

```
if (age >= 18) {
  console.log("Adult");
} else {
  console.log("Minor");
}
```

Branching with `else if`

Chain multiple conditions to handle several possible outcomes.

```
if (score >= 90) {
 grade = "A";
} else if (score >= 75) {
 grade = "B";
} else {
 grade = "C";
}
```

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